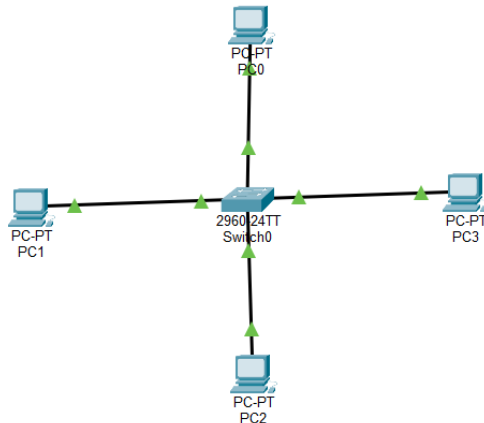


Network Topologies

1. Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.

1. Star Topology



Real-world example:

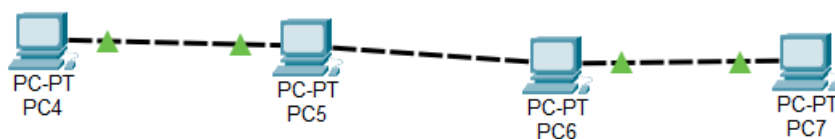
WhatsApp

A star-like network can be used in local networks where multiple devices connect through a central switch/router. The central device manages communication between the connected devices.

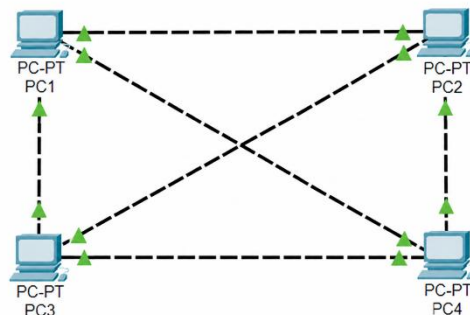
2. Bus Topology

Real-world example: Older Ethernet networks

Bus topology was used in older Ethernet networks where multiple computers shared one main communication cable.



3. Mesh Topology



Real-world example: IPL streaming

Large streaming services can use interconnected servers and network paths so data can travel through different routes. This helps maintain communication even if one network path has a problem.

2.Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

Topology		Cons	Best Use Case	App Example
Star	1. Easy to install and manage.2. Easy to find faults.3. Failure of one device does not affect others.4. Easy to add or remove devices.	1. Depends on the central switch/router.2. More cable is required.3. Central device failure can stop the network.4. Installation cost can be higher.	Home, college, and office LANs	Instagram – Users' devices connect through a central Wi-Fi router before accessing Instagram servers.
Bus	1. Simple design.2. Requires less cable.3. Low installation cost.4. Easy to set up for small networks.	1. Main cable failure can stop the network.2. Difficult to troubleshoot.3. Performance decreases with more devices.4. Limited cable length and number of devices.	Small or older LANs	Zomato – A bus-style network could be used for a small local office network where computers share one communication line.
Ring	1. Data travels in an organized manner.2. Each device gets an opportunity to transmit.3. Can provide predictable network performance.4. Less chance of data collision in traditional ring networks.	1. Failure of one connection can affect communication.2. Adding or removing devices can be difficult.3. Troubleshooting can be complicated.4. Data may need to pass through several devices.	Specialized or older LANs	Paytm – A ring-style network could be used in a controlled internal network where devices communicate in an organized sequence.
Mesh	1. Very reliable.2. Multiple paths are available.3. Failure of one connection does not necessarily stop communication.4. Provides good redundancy.	1. Requires a lot of cables/connections.2. Expensive to install.3. Difficult to configure and maintain.4. Uses more network hardware and resources.	Large, critical, and high-availability networks	Instagram – Large-scale services can use interconnected network infrastructure with multiple paths so data can continue .

3. Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

For a multiplayer mobile game like **PUBG or Free Fire**, I would choose a **Mesh topology** for the game servers. It provides multiple paths between servers, which improves **speed and reliability** because traffic can use another path if one connection fails. However, mesh networks require more hardware and connections, so they are **more expensive** to build and maintain. This makes mesh suitable for gaming systems where low delay and high reliability are important.

4. Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines.

A **Star topology** is suitable for a small warehouse because it is **cost-effective, easy to troubleshoot, and simple to manage**. If one computer or cable fails, the other computers can continue working, and more computers can easily be added later by connecting them to the switch.

